

Neural Network–Swarm Intelligence Hybrid Nonlinear Optimization Algorithm for Pavement Moduli Back-Calculation

Kasthurirangan Gopalakrishnan, Ph.D., A.M.ASCE¹

Abstract: This paper describes a novel hybrid intelligent system approach to inversion of nondestructive pavement deflection data and back-calculation of nonlinear stress-dependent pavement layer moduli. Particle swarm optimization (PSO), a population-based stochastic optimization technique inspired by social behavior of bird flocking or fish schooling, is fast emerging as an innovative and powerful computational metaphor for solving complex problems in design, optimization, control, management, business, and finance. Back-calculation of pavement layer moduli is an ill-posed inverse engineering problem which involves searching for the optimal combination of pavement layer stiffness solutions in an unsmooth, multimodal, complex search space. PSO is especially considered a robust and efficient approach for global optimization of multimodal functions. The hybrid back-calculation system described in this paper integrates finite element modeling, neural networks, and PSO in an efficient manner to mitigate the limitations and take advantages of the strengths to produce a system that is more effective and powerful than those which could be built with single technique. This is the first time the PSO approach is applied to real-time nondestructive evaluation of pavement systems.

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Introduction

The falling weight deflectometer (FWD) is a nondestructive test (NDT) device (see Fig. 1) used by pavement engineers to evaluate the structural condition of roads and to determine the moduli or stiffness of pavement layers. During FWD testing, a dynamic load is generated by a mass free falling onto a set of rubber springs and the device is set up to strike the pavement at a given force. Sensors placed around the plate and in a straight line radiating from the plate record the deflections in the pavement (analogous to ripples in a pond) induced by the falling weight. Thus, the FWD is an impulse-type testing device that imparts a transient load on the pavement surface, and the duration and magnitude of the force applied are representatives of the load pulse induced by a truck moving at moderate speeds. FWD devices are typically used to evaluate the load-bearing capacity of existing pavements and provide material properties of in situ pavement and subgrade layers for the design of pavement rehabilitation alternatives (Macdonald 2002).

In the field, the pavement deflection basins are obtained from the FWD measurements which require the use of back-calculation type structural analysis to determine pavement layer stiffnesses

and as a result estimate pavement remaining life. The accuracy of pavement strength estimation from FWD measurements is particularly important since it affects the design of pavement overlay thickness. The process of back-calculation involves comparison of FWD measured deflections with computed deflections (using a pavement response model) through an iterative optimization procedure to determine the representative pavement layer moduli which could have produced the measured FWD deflections. Presently, elastic layered analysis is commonly used in most commercial back-calculation software to predict the layer moduli typically using a deflection basin-matching approach. In this approach, layer moduli are initially assumed and theoretical surface deflections are computed. Through a series of iterations, the layer moduli are changed, and the computed deflections are then compared to the measured deflections until a match is obtained within tolerance limits. This approach has several disadvantages and does not result in unique layer moduli values as there can be more than one combination of layer moduli to achieve the match between theoretical and measured surface deflections.

Over the years, several static, dynamic, and adaptive techniques (Hoffman and Thompson 1982; Sivanesarwan et al. 1991; Uzan 1994; Meier and Rix 1995; Fwa et al. 1997; Reddy et al. 2002; Terzi et al. 2003; Gopalakrishnan 2004; Goktepe et al. 2006; Ceylan et al. 2007; Alkasawneh 2007) have been proposed for back-calculation of flexible pavement layer moduli such as the least-squares (parameter identification), database search, soft computing techniques such as neural networks (NNs), neurofuzzy systems, and genetic algorithms (GAs). Particle swarm optimization (PSO), a type of artificial intelligence method based on the collective behavior of decentralized self-organized systems, has been proved to be an efficient method for many global optimization problems and in some cases it does not suffer the difficulties

¹Research Assistant Professor, Dept. of Civil Engineering, Iowa State Univ., 354 Town Engineering Building, Ames, IA 50011. E-mail: rangan@iastate.edu

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Fig. 1. FWD attached to a van

encountered by other evolutionary computation techniques. This paper proposes a hybrid NN-PSO heuristic optimization algorithm for back-calculation of pavement layer moduli.

PSO

The PSO concept introduced by Kennedy and Eberhart (1995) draws its roots from artificial life, bird flocking, fish schooling, and swarming theory, as well as GAs and evolutionary programming. Although it was originally introduced for optimization of nonlinear continuous functions, many advances in PSO development have enabled it to handle a wide class of complex engineering and science optimization problems. Similar to GAs, a population of potential solutions to the problem under consideration is used to probe the search space in PSO. However, each individual of the population in PSO has an adaptable *velocity* (position change) according to which it moves in the search space. Moreover, each individual has a *memory*, remembering the best position of the search space it has ever visited (Eberhart et al. 1996). The movement of the individual is thus an aggregated acceleration toward its best previously visited position and toward the best individual of a topological neighborhood. Since the “acceleration” term was mainly used for particle systems in particle physics (Reeves 1983) and the term “swarm” for describing population, this algorithm was named as PSO. In essence, PSO employs a swarm of particles or possible solutions that fly through the feasible solution space to explore optimal solutions.

Two variants of the PSO algorithm were developed: one with a global neighborhood (Gbest model) and one with a local neighborhood. Each particle moves toward its best previous position and toward the best particle in the whole swarm, according to the global variant. In contrast, each particle moves toward its best previous position and toward the best particle in its restricted neighborhood, according to the local variant (Eberhart et al. 1996).

The global variant PSO algorithm, which is the most standard one, is described as follows. Suppose that the search space is D dimensional, then the i th particle of the swarm can be represented by a D -dimensional vector, $\mathbf{X}_i = (x_{i1}, x_{i2}, \dots, x_{iD})^T$. The velocity (position change) of this particle can be represented by another D -dimensional vector $\mathbf{V}_i = (v_{i1}, v_{i2}, \dots, v_{iD})^T$. The best previously visited position of the i th particle is denoted as $P_i = (p_{i1}, p_{i2}, \dots, p_{iD})^T$. Define g as the index of the best particle in the swarm (i.e., the g th particle is the best) and let the superscripts

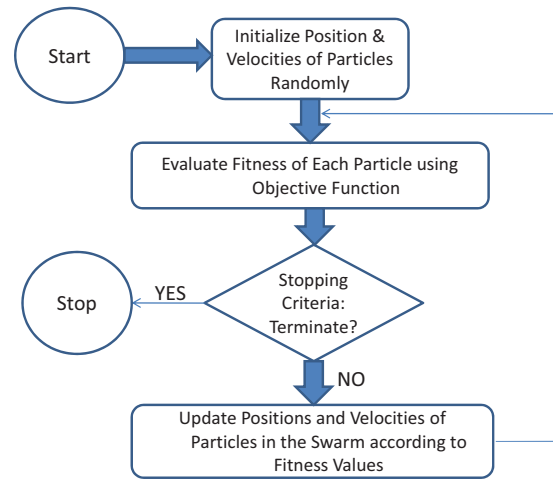


Fig. 2. Standard PSO algorithm flowchart

denote the iteration number. Each particle updates its position based on its own best exploration, best swarm overall experience, and its previous velocity vector according to the following two equations (Eberhart et al. 1996) which define the initial version of the PSO algorithm:

$$v_{id}^{n+1} = v_{id}^n + cr_1^n(p_{id}^n - x_{id}^n) + cr_2^n(p_{gd}^n - x_{id}^n) \quad (1)$$

$$x_{id}^{n+1} = x_{id}^d + v_{id}^{n+1} \quad (2)$$

where $d = 1, 2, \dots, D$; $i = 1, 2, \dots, N$, where $N = \text{size of the swarm}$; $c = \text{positive constant, called acceleration constant}$; r_1 and $r_2 = \text{random numbers, uniformly distributed in } [0, 1]$; and $n = 1, 2, \dots$ determines the iteration number. The performance of each particle is measured according to a predefined fitness function or objective function, which is related to the problem under consideration.

Later on, a maximum velocity parameter, V_{max} , was imposed to improve the efficiency of PSO in the region of the optimum by allowing its velocity step size to continue the search at a finer grain. Thus, in the later versions of the PSO, Eqs. (1) and (2) were modified as follows by incorporating a weight parameter for the previous velocity of the particle (Eberhart and Shi 1998):

$$v_{id}^{n+1} = \chi[wv_{id}^n + c_1r_1^n(p_{id}^n - x_{id}^n) + c_2r_2^n(p_{gd}^n - x_{id}^n)] \quad (3)$$

$$x_{id}^{n+1} = x_{id}^d + v_{id}^{n+1} \quad (4)$$

where w is called *inertia weight*; $c_1, c_2 = \text{two positive constants, called cognitive and social parameters, respectively}$; and $\chi = \text{constriction factor}$, which is used alternatively to w to limit velocity (Parsopoulos and Vrahatis 2002). The swarm behavior in conventional PSO is influenced by the number of particles, inertia weight, maximum velocity, and acceleration coefficients to modify the velocity. A simple flowchart for the standard PSO algorithm is illustrated in Fig. 2.

The choice of the PSO algorithm's parameters is considered to be of utmost importance for the speed, convergence, and efficiency of the algorithm. The impact of the previous history of velocities on the current velocity is controlled by employing the inertia weight, w , which influences the trade-off between global (wide-ranging) and local (nearby) exploration abilities of the “flying points.” The global exploration (searching new areas) capability is facilitated by a larger inertia weight, while a smaller inertia weight tends to facilitate local exploration to fine tune the

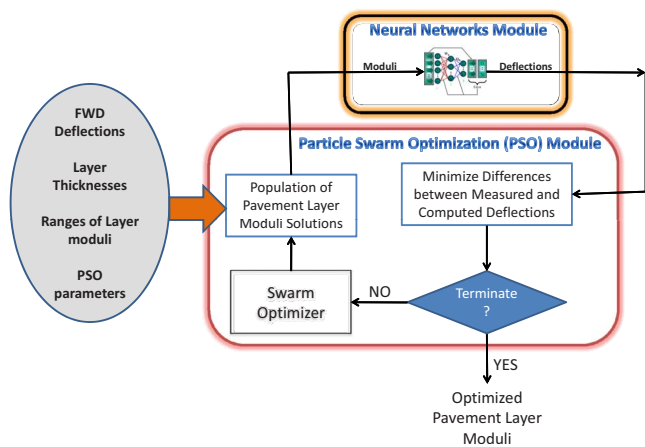


Fig. 3. NN-PSO hybrid pavement moduli back-calculation approach

current search area. Thus, a good balance between global and local exploration abilities requiring less iterations on average to find the optimum solution can be achieved by suitable selection of the inertia weight (Shi and Eberhart 1998).

It has been shown that PSO adheres to five basic principles of swarm intelligence outlined by Eberhart et al. (1996): (1) proximity; (2) quality; (3) diverse response; (4) stability; and (5) adaptability. PSO requires only primitive mathematical operators as it involves only two model equations and is computationally inexpensive in terms of both memory requirements and speed. It has several advantages over other optimization techniques (Al-rashidi and El-Hawary 2006): its derivative-free nature unlike many conventional techniques, flexibility to be integrated with other optimization and soft computing techniques to form hybrid tools, fewer parameters to adjust, ability to handle objective functions with stochastic nature, ease of implementation, lack of requirement of seed solution to start its iteration process, etc. These advantages make the PSO algorithm a promising alternative for handling the problem of pavement moduli back-calculation which is a global optimization problem wherein the objective is to determine the unknown pavement layer moduli that minimize the difference between measured and computed deflections. Although GAs have been successfully used in the past for pavement moduli back-calculation, some of the significant advantages of PSO over GAs (easy to implement, few parameters to adjust, faster convergence to best solution, etc.) provided the impetus for exploring the use of PSO for solving the back-calculation problem in this study.

Implementation of Neuroswarm Intelligence Hybrid Back-Calculation Tool

Overall Framework

The objective (fitness) function or the cost function for the proposed NN-PSO hybrid optimization approach is the difference between measured FWD deflections and computed pavement surface deflections. In this paper, the implementation of the NN-PSO optimization approach is discussed for a three-layered flexible pavement structure although it can be used for other pavement types with varying number of layers owing to its flexible and integrated modular system approach. A typical three-layered flexible pavement structure consists of hot-mix asphalt (HMA) sur-

face layer, a granular base layer consisting of unbound aggregates, and the bottommost layer consisting of subgrade soils.

In the proposed NN-GA hybrid optimization approach (see Fig. 3), a trained NN serves as a surrogate forward pavement response model that has learned the mapping between pavement layer elastic moduli and resulting pavement surface deflections for a variety of case scenarios generated using a two-dimensional (2D) axisymmetric pavement finite-element (FE) program (Raad and Figueroa 1980). The PSO, in essence, finds the optimal values of the NN inputs (pavement layer moduli) iteratively such that the corresponding values of the network outputs (deflections) match the measured pavement surface deflections to minimize the differences between the measured and computer deflections. Although the error-minimization deflection-based objective function can be defined in a number of ways, a simple objective function representing sum of the squared differences between measured and computed deflections as shown in Eq. (5) was selected for this study (where $n=6$)

$$f = \sum_{i=1}^n (D_i - d_i)^2 \quad (5)$$

The NN-PSO optimization framework was implemented in MATLAB. The input variables include six FWD measured surface deflections at 300-mm radial offsets starting from the center of the FWD loading plate, the HMA surface and base layer thicknesses, and the corresponding min-max ranges of pavement layer moduli. As mentioned previously, the choice of the PSO algorithm's parameters is crucial in achieving convergence of solution for the problem under consideration. So far, there have not been any reported studies on the use of PSO algorithm for back-calculation of pavement moduli and the current study is considered to be the first attempt. The PSO parameters were selected based on guidance from previous studies reported on other applications (Shi and Eberhart 1998; Schutte and Groenword 2005). The swarm (population) size was set to 25 and the cognitive acceleration coefficient and social acceleration coefficient were set to 2.8 and 1.3, respectively. The constriction factor was set to 1.0 (Clerc 1999).

NNs Module

The elastic layered programs (ELPs) used in the analysis and design of flexible pavements consider the pavement as an elastic multilayered media and assume that pavement materials are linear elastic, homogeneous, and isotropic. However, the unbound granular materials and fine-grained subgrade soils, referred to as pavement geomaterials, do not follow a linear stress-strain behavior under repeated traffic loading. The nonlinearity or stress dependency of resilient modulus for unbound granular materials and cohesive fine-grained subgrade soils has been well established (Brown and Pappin 1981; Thompson and Elliot 1985; Garg et al. 1998). Unbound aggregates exhibit stress hardening type behavior, whereas fine-grained subgrade soils show stress-softening type behavior.

Over the years, NNs have emerged as successful computational tools for studying a majority of pavement engineering problems (Meier and Rix, 1995; Ceylan et al. 2007). In the development of the mechanistic-empirical pavement design guide for the American Association of State Highway and Transportation Officials, NNs have been recognized as nontraditional yet very powerful computing techniques and were used in preparing

the concrete pavement analysis package. Recent research studies at the Iowa State University have focused on the development of NN-based forward and back-calculation type highway flexible pavement analysis models to predict critical pavement responses and layer moduli, respectively (Ceylan et al. 2007).

The basic element in NNs is a processing element (artificial neuron). An artificial neuron receives information (signal) from other neurons, processes it, and then relays the filtered signal to the other neurons (Tsoukalas and Uhrig 1997). The receiving end of the neuron has incoming signals $X_1, X_2, \dots, X_n$. Each of them is assigned a weight, which is given based on experience and which may change during the training process. The summation of all the weighted signal amounts yields the combined input quantity I_k . The combined input quantity I_k is then sent to a preselected transfer function (sometimes called an activation function) T , and a filtered output Y_k is generated in the outgoing end of the artificial neuron k through the mapping of the transfer function. The process can be written in the form of the following equations:

$$I_k = \sum_{i=1}^n w_{ik} x_i \quad (6)$$

$$T_k = T(I) \quad (7)$$

There are several types of transfer functions that can be used, including sigmoid, threshold, and Gaussian functions. The transfer function most often used is the sigmoid function because of its differentiability. The sigmoid function can be represented by the following equation:

$$T(I) = \frac{1}{1 + \exp(-\varphi I)} \quad (8)$$

where φ =positive scaling constant, which controls the steepness between the two asymptotic values 0 and 1 (Tsoukalas and Uhrig 1997). The *back-propagation* (BP) learning algorithm is the most commonly used NN training algorithm in which the network learns the relationship between stipulated input-output data pairs in a supervised manner. In the BP learning algorithm, the error energy used for monitoring the progress toward convergence is the generalized value of all errors that is calculated by the least-squares formulation and represented by a mean squared error (MSE) as follows (Haykin 1999):

$$MSE = \frac{1}{MP} \sum_{i=1}^P \sum_{k=1}^M (d_k - y_k)^2 \quad (9)$$

where y_k and d_k =actual and desired outputs, respectively; M =number of neurons in the output layer; and P represents the total number of training patterns. Other performance measures such as the root MSE, average absolute error (AAE), etc., are also used.

Thus, a multilayer feedforward NNs with BP learning algorithm was employed in developing a surrogate forward calculation procedure to map the relation between input layer thicknesses and moduli and output surface deflections for passing on to the PSO module during fitness evaluation. The goal is to simulate heavy weight deflectometer (HWD) loading using a numerical model for a wide variety of layer thicknesses and combinations of layer moduli encountered in the field resulting in a comprehensive synthetic solution set. A 2D axisymmetric FE program (Raad and Figueroa 1980) developed at the University of Illinois and commonly used in the structural analysis of flexible pavements was employed to generate a comprehensive synthetic database of moduli-deflection solutions for wide ranges of layer thicknesses

and layer moduli. Numerous research studies have validated that the FE model used in this study (Raad and Figueroa 1980) provides a realistic pavement structural response prediction for both highway and airfield pavements by incorporating stress-sensitive geomaterial models, the typical hardening behavior of nonlinear unbound aggregate bases and softening nature of subgrade soils under increasing stress states, and the Mohr-Coulomb failure criteria to limit material strength.

The synthetic database of FE solutions constituted the training and testing sets for developing NN-based models for rapid forward analysis of flexible pavements. A generic three-layer flexible pavement structure consisting of HMA surface layer, unbound aggregate base layer, and subgrade layer was modeled using the FE software. The top surface HMA layer was characterized as a linear-elastic material with Young's modulus, E_{HMA} , and Poisson's ratio, ν . The K - θ model (Hicks and Monismith 1971) was used as the nonlinear characterization model for the unbound aggregate layer

$$E_R = K\theta^n \quad (10)$$

where E_R =resilient modulus (MPa); $\theta = \sigma_1 + \sigma_2 + \sigma_3 = \sigma_1 + 2\sigma_3$ =bulk stress; and K and n =multiple regression constants obtained from repeated load triaxial test data on granular materials. Based on the work reported by Rada and Witczak (1981), K and n model parameters can be correlated to characterize the nonlinear stress-dependent behavior with only one model parameter. Thus, good quality granular materials, such as crushed stone, show higher K and lower n values, whereas the opposite applies for lower quality aggregates.

Fine-grained subgrade soils were modeled using the commonly used bilinear resilient modulus model (Thompson and Elliot 1985)

$$E_R = E_{Ri} + K_1(\sigma_d - \sigma_{di}) \quad \text{for } \sigma_d < \sigma_{di}$$

$$E_R = E_{Ri} + K_2(\sigma_d - \sigma_{di}) \quad \text{for } \sigma_d > \sigma_{di} \quad (11)$$

where E_{Ri} ="breakpoint" resilient modulus; σ_d =breakpoint deviator stress ($\sigma_d = \sigma_1 - \sigma_3$); σ_{di} =breakpoint deviator stress; and K_1 and K_2 =statistically determined coefficients from laboratory tests. E_{Ri} can be used to classify fine-grained subgrade soils as being soft, medium, or stiff. Based on extensive repeated laboratory testing data at the University of Illinois, Thompson and Robnett (1979) indicated that E_{Ri} , typically associated with a repeated deviator stress of about 41 kPa (6 psi), is a good indicator of the subgrade soil's resilient modulus. Also, the Asphalt Institute's thickness design manual MS-1 (The Asphalt Institute 1982) recommends E_{Ri} as the subgrade modulus input for ELP analysis.

Thus, HMA modulus, E_{HMA} , granular base K - θ model parameter, K , and subgrade break-point resilient moduli, E_{Ri} , were used as the layer stiffness inputs for all the FE runs. The 40-kN wheel load was applied as a uniform pressure of 552 kPa over a circular area of radius 150 mm simulating the HWD loading. A comprehensive FE synthetic database was generated by varying the HMA layer thickness (in the range of 75–400 mm), aggregate base layer thickness (in the range of 100–550 mm), E_{HMA} (in the range of 6.9–41.5 GPa), K (in the range of 21–82 MPa), and E_{Ri} (in the range of 7–105 MPa) for NN training and testing. The ranges of pavement layer properties used in developing the FE synthetic database are summarized in Table 1. Independent data sets were used for NN training and testing. In this study, the 5-40-40-6 architecture (five inputs, two hidden layers with 40 neurons each, and six outputs) was chosen as the best architecture for the NN forward model based on its lowest training and testing

Table 1. Ranges of Pavement Layer Properties Used in Developing the FE Synthetic Database

Material type	Layer thickness	Material model	Layer modulus inputs	Poisson's ratio
Asphalt concrete	$h_{AC}=75\text{--}400$ mm (3–15 in.)	Linear elastic	$E_{AC}=690\text{--}41,500$ MPa (100–6,000 ksi)	$\nu=0.35$
Unbound aggregate base	$h_{GB}=102\text{--}550$ mm (4–22 in.)	Nonlinear K - θ model	$M_R=K\theta^n$; $K=20.7\text{--}82.8$ MPa (3–12 ksi)	$\nu=0.35$ for $K\geq 34.5$ MPa (5 ksi); $\nu=0.40$ for $K<34.5$ MPa (5 ksi)
Fine-grained subgrade	7,620 mm (300 in.)—total pavement thickness	Nonlinear bilinear model	$M_R=f(E_{Ri})$; $E_{Ri}=6.9\text{--}105$ MPa (1–15 ksi)	$\nu=0.45$

MSEs for all six deflection output variables. The five inputs correspond to three-layer stiffnesses, HMA layer thickness, and base layer thickness. AAEs were calculated as sum of the individual absolute relative errors divided by the number of independent testing patterns

$$AAE (\%) = \sum_{i=1}^n \left| \frac{y_{\text{actual}} - y_{\text{predicted}}}{y_{\text{actual}}} \right| \times 100 \quad (12)$$

where i = i th testing pattern among n testing patterns. The AAE values for NN predicted output deflections were in the range of 0.2–0.4% with R^2 values above 0.999, indicating proper training and excellent prediction performance of the NN surrogate forward calculation model.

Back-Calculation of Pavement Moduli Using Neuroswarm Optimization Approach

Hypothetical Data

Hypothetical data covering wide ranges of layer thicknesses and FWD deflections commonly encountered in the field were first used to evaluate the prediction accuracy of the developed NN-PSO hybrid back-calculation tool. A total of about 150 data sets was independently selected from the comprehensive synthetic FE solutions database to assess the prediction performance. The performance of NN-PSO optimization approach in back-calculating nonlinear flexible pavement layer moduli (E_{HMA} and E_{Ri}) is reported in Figs. 4 and 5. As shown in Figs. 4 and 5, all 150 neuroswarm based back-calculation predictions fell on the line of

equality for the two pavement layer moduli thus indicating a proper training and very good performance of the proposed hybrid back-calculation model. However, the base layer moduli predictions (K or n), not reported here, were as such poor due to well-known problems associated with back-calculating base layer moduli that have been recognized by previous research studies (Ceylan et al. 2007). It has been proposed that by including the predicted HMA and subgrade moduli among the inputs, the base moduli can be successfully predicted separately.

In Figs. 6 and 7, the trajectory of the particle swarm's flight through the search space is presented for two extreme cases (low and high moduli). The PSO's ability for speedy convergence to global minima is demonstrated through these two extreme examples.

Case Study 1: Field Data

Next, the hybrid NN-PSO back-calculation tool was validated using actual field data acquired from an airport flexible pavement test section referred to as conventional flexible pavement with medium-strength subgrade (MFC) at the U.S. National Airport Pavement Test Facility (NAPTF). MFC is a conventional granular base flexible pavement resting over a medium-strength subgrade. It consists of 127-mm (5-in.)-thick HMA surface course, 200-mm-thick crushed stone granular base, and 307-mm-thick granular sub-base on top of the subgrade. A CL-CH soil classification (ASTM Unified Soil Classification System), material known as Dupont clay, was used for the medium-strength subgrade (target California bearing ratio of 8). The naturally occurring sandy-soil material (SW-SM soil classification) at the full-scale test site underlies the subgrade layer. Detailed information related to NAPTF

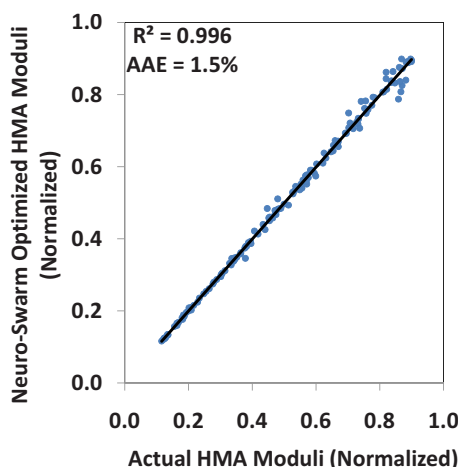


Fig. 4. Prediction performance of NN-PSO hybrid back-calculation tool—HMA moduli

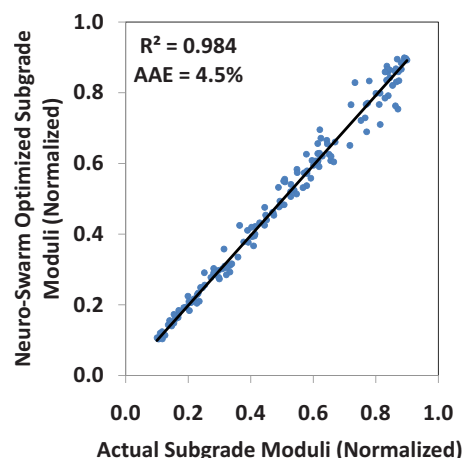


Fig. 5. Prediction performance of NN-PSO hybrid back-calculation tool—subgrade moduli

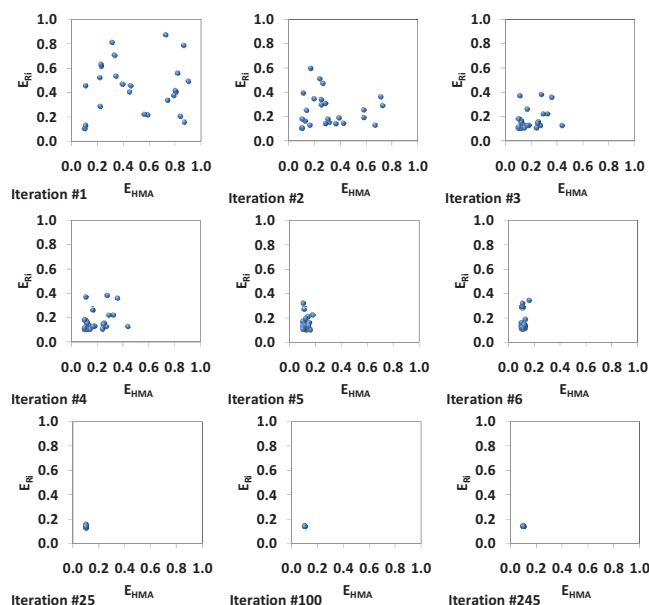


Fig. 6. Particle swarm's flight through (normalized) moduli search space—Case 1 (actual solution): $E_{HMA}=0.100$; $E_{Ri}=0.147$

flexible test sections, material properties, and analysis of NDT data are found in Gopalakrishnan (2004). The MFC NDT data referenced in this paper are accessible for download at the Federal Aviation Administration (FAA) Airport Technology website (www.airporttech.tc.faa.gov).

NDTs using the FWD equipment were conducted on MFC section prior to traffic testing to verify the uniformity of pavement and subgrade construction and strength. The FWD equipment was equipped with a segmented 30.5-cm (12-in.) loading plate. A pulse width of 27–30 ms was used during the FWD tests. Surface deflection basins from FWD tests conducted on June 14, 1999 (pavement temperature=21.2°C) at nominal force amplitudes of 40 kN (9,000 lb) were used in this study. A schematic of FWD

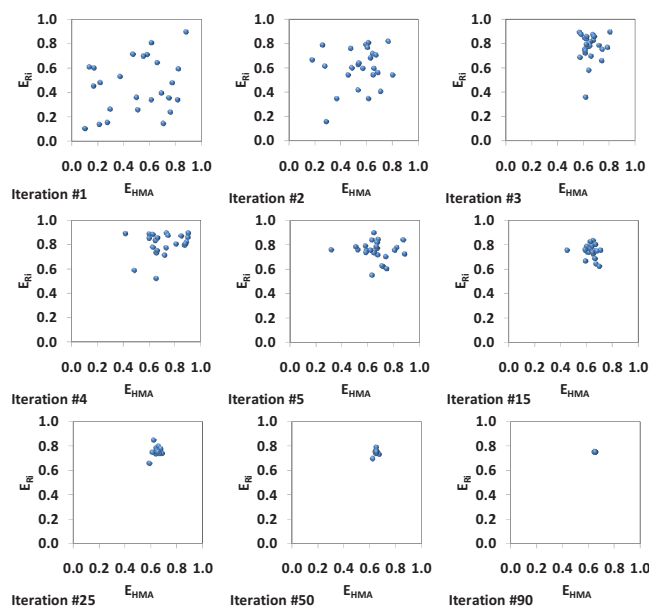


Fig. 7. Particle swarm's flight through (normalized) moduli search space—Case 2 (actual solution): $E_{HMA}=0.644$; $E_{Ri}=0.745$

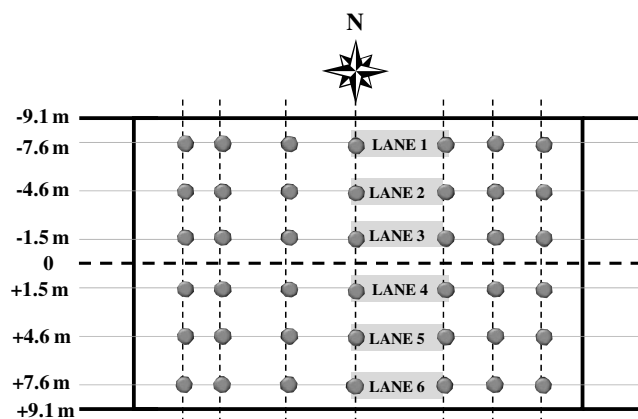


Fig. 8. FWD test locations in MFC airport flexible pavement test section

test locations in MFC section is shown in Fig. 8. A total of 42 test locations (seven stations $\times$ six lanes) was tested with the FWD device. Fig. 9 displays the variability in maximum surface deflections (D_0) in MFC section. The NN-PSO moduli predictions for MFC test section are shown in Fig. 10. As seen in Fig. 10, the predicted moduli closely follow the trend in the observed variability in the maximum FWD surface deflection, D_0 , which is an indicator of the overall stiffness of the pavement (i.e., the higher the deflection is, lower the back-calculated moduli and vice versa).

The NN-PSO back-calculation results were then compared with three existing back-calculation methods: (1) regression algorithms (Thompson 1987); (2) BAKFAA (Hayhoe 2002); and (3) WESDEF (Van Cauwelaert et al. 1989). Thompson (1987) developed direct plot procedures and regression algorithms for back-calculating pavement layer and subgrade moduli from 40-kN (9-kip) FWD test results based on a synthetic database generated using the same FE program used in this study. The ranges of pavement layer properties used in developing the database were smaller than those used in the current study. Both BAKFAA and WESDEF are ELP-based back-calculation programs.

The BAKFAA back-calculation program was developed under the sponsorship of the FAA Airport Technology Branch and was

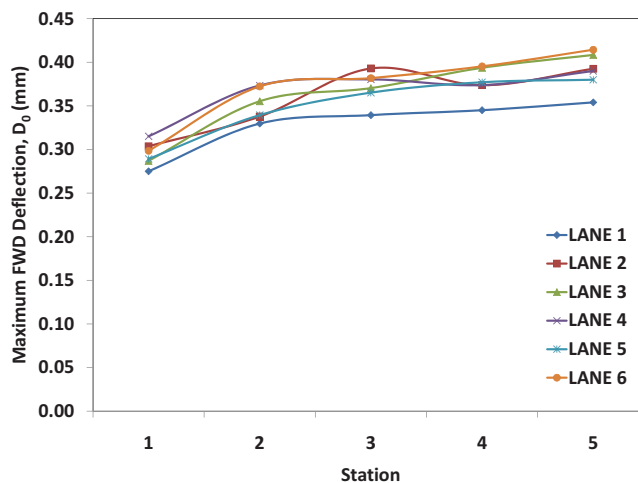


Fig. 9. Variability in FWD measured maximum surface deflections in MFC

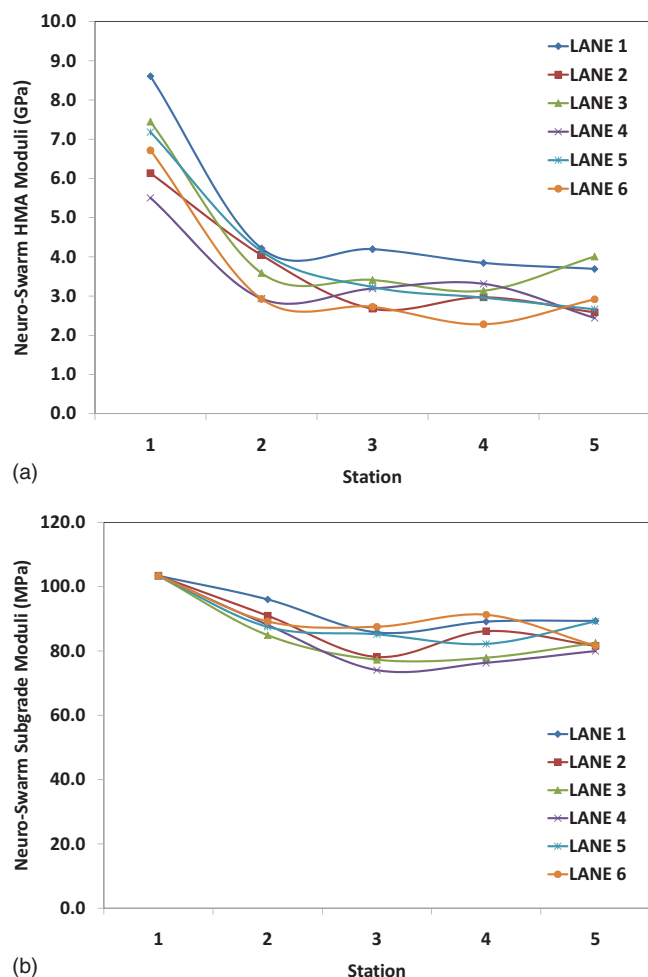


Fig. 10. Neuroswarm pavement layer moduli predictions for MFC

based on the FAA's Layered Elastic Analysis Program (LEAF) ELP (Hayhoe 2002). In this program, the pavement layer moduli and subgrade moduli are adjusted to minimize the root mean square of the differences between FWD sensor measurements and the LEAF-computed deflection basin for a specified pavement structure. A standard multidimensional simplex optimization routine is then used to adjust the moduli values (McQueen et al. 2001). In this study, a stiff layer with a modulus of 6.9 GPa (1×10^6 psi) and Poisson's ratio of 0.50 was used in back-calculation. Based on the as-constructed conditions, the stiff layer was set at 3 m (10 ft) for the MFC.

The WESDEF back-calculation program developed uses the WESLEA multilayer elastic analysis program. It utilizes an iterative procedure to obtain a set of moduli that, when used in linear-elastic calculations, will produce deflections similar to the measured values. The program has the ability to back-calculate moduli values using deflections with depth, such as those obtained using multidepth deflectometers (MDDs), as well as with surface deflections. The material type, entered for each layer in the pavement structure, is used to establish the default initial modulus, minimum and maximum moduli, Poisson's ratio, and the interface slip values. The modulus for the stiff layer was set to 6.9 GPa (1×10^6 psi) with Poisson's ratio of 0.50. A Visual Basic program was written to enable WESDEF to handle batch computations.

The NN-PSO back-calculated flexible pavement layer moduli for MFC are plotted against those predicted by the other three

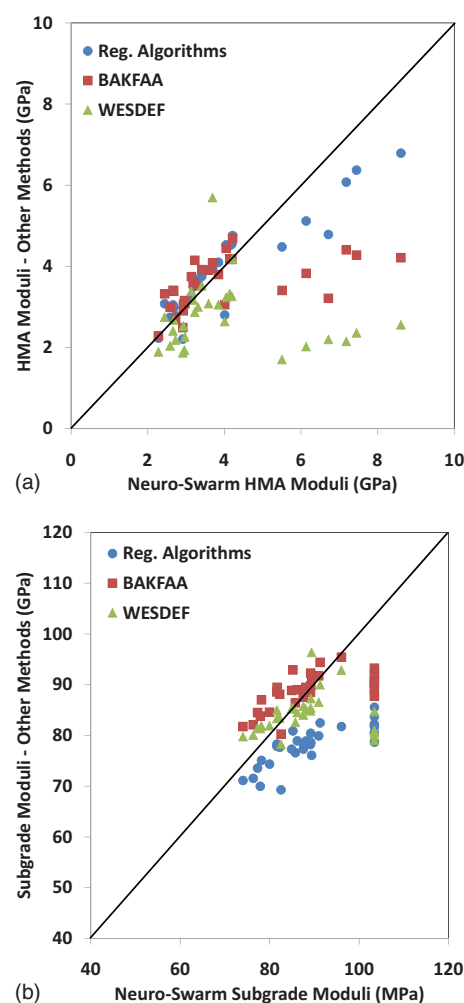


Fig. 11. Comparison of neuroswarm predictions with the other back-calculation methods

methods in Fig. 11. In general, the neuroswarm moduli predictions are consistent and agreeable with those predicted by the existing approaches. In the case of HMA moduli, the neuroswarm predictions are very consistent with the other three methods in the lower bracket while the moduli values are overpredicted in the upper end compared to the other three methods. Note that the ELP-based back-calculation programs, BAKFAA and WESDEF, assume the subgrade to be linear elastic while both the regression algorithms and the NN-PSO prediction methodology consider the nonlinear stress-dependent subgrade properties. However, the

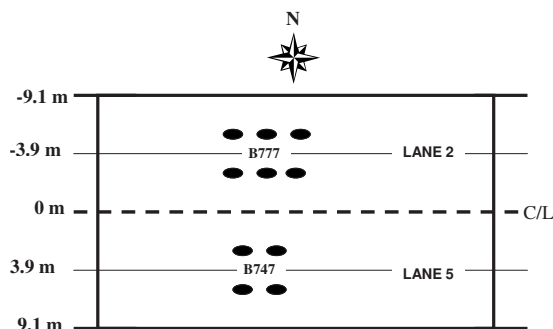


Fig. 12. HWD test lane locations during NAPTf traffic testing

NN-PSO methodology is based on a population-based stochastic optimization technique which is different from the approach employed by all the other three methods (Fig. 12).

Case Study 2: Field Data

In the previous case study, the MFC test section (a conventional granular base flexible pavement) was considered. In this case study involving accelerated pavement testing, the MFS test section (an asphalt-stabilized base flexible pavement) was considered which consisted of 127 mm (5 in.) of P-401 HMA surface layer, 127 mm (5 in.) of P-401 HMA stabilized base layer (treated together as 254 mm of HMA layer on top), and 216 mm (8.5 in.) of P-209 granular base/sub-base layer and a medium-strength subgrade (same as MFC).

The first series of NAPTF traffic tests, referred to as CC-1 traffic testing, began in February 2000 and was completed in September 2001. During NAPTF CC-1 traffic testing, a six-wheel dual-tridem (B777) landing gear, with 1,372-mm (54-in.) dual spacing and 1,448-mm (57-in.) tandem spacing, was loaded on LANE 2 (see Fig. 8) while LANE 5 (see Fig. 8) was loaded with a four-wheel dual-tandem (B747) landing gear having 1,118-mm (44-in.) dual spacing and 1,473-mm (58-in.) tandem spacing. The wheel loads were set to 20,412 kg (45,000 lb) each and the tire pressure was 1,295 kPa (188 psi). The traffic speed was 8 km/h (5 mph) throughout the traffic test program. To realistically simulate transverse aircraft movements, a wander pattern consisting of a fixed sequence of 66 vehicle passes (33 traveling in the east direction and 33 traveling in the west direction), arranged in nine equally spaced wander positions (or tracks) at intervals of 260 mm (10.25 in.), was used during traffic testing.

The NAPTF failure criterion was the one established in the U.S. Army Corps of Engineers' multiwheel heavy gear load tests (Ahlvin et al. 1971). Failure is defined as the presence of at least 25.4-mm (1-in.) surface upheaval adjacent to the traffic lane. This is linked to a structural or shearing failure in the subgrade. The NAPTF test sections were trafficked until they were deemed failed.

NDTs using a HWD impulse device were conducted to monitor the effect of full-scale trafficking on pavement response and performance over time. The FAA HWD equipment was also configured with a 30.5-cm (12-in.) loading plate and a pulse width of 27–30 ms was used. The HWD tests were conducted at nominal force amplitudes of 53.4 (12,000), 106.7 (24,000), and 160 kN (36,000 lb). The location and orientation of HWD test lanes are illustrated in Fig. 6. The HWD tests were performed on the untrafficked centerline (CL) and on the B777 and B747 traffic lanes (LANE 2 and LANE 5). The HWD test sequences were repeated at 3-m (10-ft) intervals along the test lanes.

The comparative effects of repeated aircraft trafficking on airfield pavement layer moduli, especially the HMA layer moduli and subgrade moduli, were studied by applying the hybrid NN-PSO back-calculation methodology on NAPTF flexible pavement HWD data acquired during full-scale traffic testing. Fig. 13 captures the structural deterioration in MFS test section via changes in NN-PSO back-calculated HMA moduli (E_{HMA}) and nonlinear stress-dependent subgrade moduli (E_{Ri}) with the number of traffic load repetitions (N). The results for the B777 and B747 traffic lanes are plotted together with results for the untrafficked CL where the changes in moduli values were mainly due to seasonal and temperature variations. Not only the proposed back-calculation approach captures the relative severity of B777 and B747 aircraft loadings on pavement layer moduli, but it also cap-

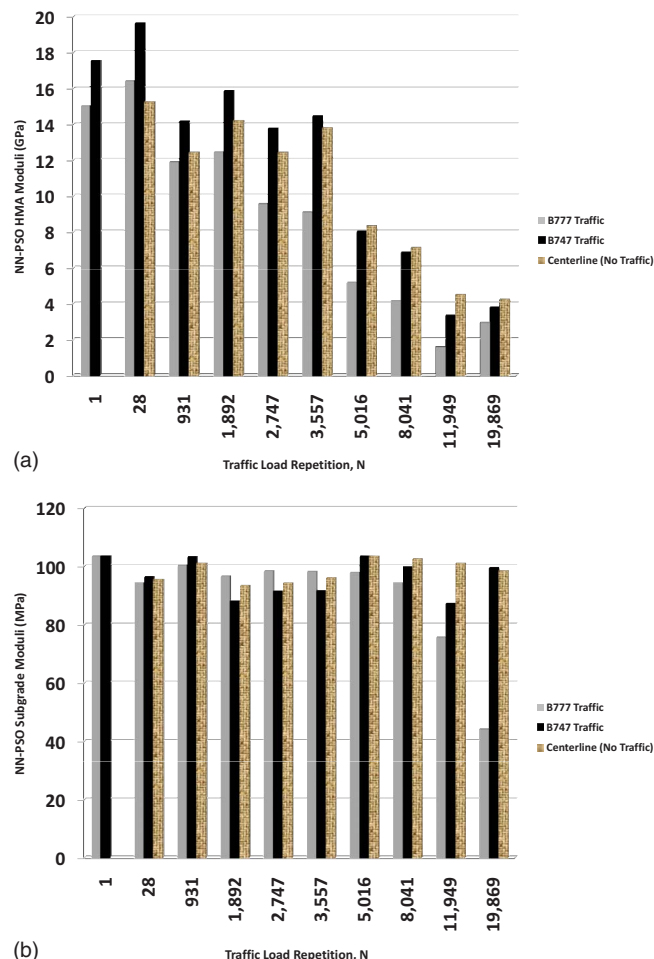


Fig. 13. Effect of full-scale trafficking on NN-PSO back-calculated moduli in MFS test section: (a) HMA moduli; (b) nonlinear subgrade moduli

tures the impact of repeated aircraft loading and “no loading” on airfield pavements in general. The back-calculation trends for the MFS section are consistent with the MDD results and HMA strain gauge results as well as trench study findings reported by Gopalakrishnan (2004). These results demonstrate the robustness and consistency of the proposed NN-PSO hybrid back-calculation approach.

Summary and Conclusions

This paper presented the development of a new hybrid NN-PSO approach for the back-calculation of flexible pavement layer moduli. For the first time, the PSO technique, which is increasingly becoming popular in engineering optimization, has been applied to inversion of pavement NDT deflection data and back-calculation of HMA layer moduli and stress-dependent nonlinear subgrade moduli. The flexibility of PSO technique to be integrated with other optimization and soft computing techniques to form hybrid tools, requiring only fewer parameters to adjust, ability to handle objective functions with stochastic nature, ease of implementation, lack of requirement of seed solution to start its iteration process, etc., makes it a very attractive alternative for handling the back-calculation problem. It was demonstrated using both hypothetical data and actual field data that the proposed hybrid NN-PSO can successfully predict the flexible pavement

surface layer moduli and nonlinear subgrade moduli. Future efforts will focus on conducting sensitivity analysis to derive the optimal PSO parameter settings for handling different field conditions (Adeli 2001).

References

- Adeli, H. (2001). "Neural networks in civil engineering: 1989–2000." *Comput. Aided Civ. Infrastruct. Eng.*, 16, 126–142.
- Ahlvin, R. G., et al. (1971). "Multiple-wheel heavy gear load pavement test." *Technical Rep. S-71-17*, Vol. I, US Army Corps of Engineers, Waterways Experiment Station, Vicksburg.
- Alkasawneh, W. (2007). "Backcalculation of pavement moduli using genetic algorithms." Ph.D. thesis, Univ. of Akron.
- Alrashidi, M., and El-Hawary, M. (2006). "A survey of particle swarm optimization applications in power system operations." *Electric Power Components and Systems*, 34(12), 1349–1357.
- The Asphalt Institute. (1982). "Research and development of The Asphalt Institute's thickness design manual (MS-1)." *Research Rep. No. 82-2*, The Asphalt Institute, College Park, Md.
- Brown, S. F., and Pappin, J. W. (1981). "Analysis of pavements with granular bases." *Transp. Res. Rec.*, 810, 17–23.
- Ceylan, H., Guclu, A., Bayrak, M. B., and Gopalakrishnan, K. (2007). "Nondestructive evaluation of Iowa pavements: Phase I." CTRE, Iowa State Univ., Ames, Iowa, CTRE Project No. 04-177.
- Clerc, M. (1999). "The swarm and the queen: Towards a deterministic and adaptive particle swarm optimization." *Proc., IEEE Congress on Evolutionary Computation*, 1951–1957.
- Eberhart, R., Simpson, P., and Dobbins, R. (1996). *Computational intelligence PC tools*, Academic, San Diego.
- Eberhart, R. C., and Shi, Y. (1998). "Evolving artificial neural networks." *Proc., 1998 Int. Conf. on Neural Networks and Brain*, Beijing, 5–13.
- Fwa, T. F., Tan, C. Y., and Chan, W. T. (1997). "Backcalculation analysis of pavement-layer moduli using genetic algorithms." *Transp. Res. Rec.*, 1570, 134–142.
- Garg, N., Tutumluer, E., and Thompson, M. R. (1998). "Structural modeling concepts for the design of airport pavements for heavy aircraft." *Proc., 5th Int. Conf. on the Bearing Capacity of Roads and Airfields*, Trondheim.
- Goktepe, A. B., Agar, E., and Lav, A. H. (2006). "Comparison of multilayer perceptron and adaptive neuro-fuzzy system on backcalculating the mechanical properties of flexible pavements." *ARI: The Bulletin of the Istanbul Technical University*, 54(3), 1–13.
- Gopalakrishnan, K. (2004). "Performance analysis of airport flexible pavement subjected to new generation aircraft." Ph.D. thesis, Univ. of Illinois at Urbana-Champaign.
- Hayhoe, G. H. (2002). "LEAF—A new layered elastic computational program for FAA pavement design and evaluation procedures." *Proc., 2002 Federal Aviation Administration Airport Technology Transfer Conf.*
- Haykin, S. (1999). *Neural networks: A comprehensive foundation*, Prentice-Hall, Englewood Cliffs, N.J.
- Hicks, R. G., and Monismith, C. L. (1971). "Factors influencing the resilient properties of granular materials." *Transp. Res. Rec.*, 345, 15–31.
- Hoffman, M. S., and Thompson, M. R. (1982). "Backcalculating nonlinear resilient moduli from deflection data." *Transp. Res. Rec.*, 852, 42–51.
- Kennedy, J., and Eberhart, R. C. (1995). "Particle swarm optimization." *Proc., IEEE Int. Conf. on Neural Networks*, Vol. 4, Piscataway, N.J., 1942–1948.
- Macdonald, R. A. (2002). "Pavement technology project in Thailand." *Road Directorate Rep. No. 119*, Danish Road Institute, Denmark.
- McQueen, R. D., Marsey, W., and Arze, J. M. (2001). "Analysis of non-destructive data on flexible pavement acquired at the national airport pavement test facility." *Proc., Airfield Pavement Specialty Conf.*, ASCE, Chicago.
- Meier, R. W., and Rix, G. J. (1995). "Backcalculation of flexible pavement moduli from dynamic deflection basins using artificial neural networks." *Transp. Res. Rec.*, 1473, 72–81.
- Parsopoulos, K. E., and Vrahatis, M. N. (2002). "Recent approaches to global optimization problems through particle swarm optimization." *Nat. Comput.*, 1, 235–306.
- Raad, L., and Figueroa, J. L. (1980). "Load response of transportation support systems." *Transp. Engrg. J.*, 106(TE1), 111–128.
- Rada, G., and Witczak, M. W. (1981). "Comprehensive evaluation of laboratory resilient moduli results for granular material." *Transportation Research Record. 810*, TRB, National Research Council, Washington, D.C.
- Reddy, M. A., Reddy, K. S., and Pandey, B. B. (2002). "Backcalculation of pavement moduli using genetic algorithms." *J. Highw. Res. Board*, 66, 1–10.
- Reeves, W. T. (1983). "Particle systems—A technique for modelling a class of fuzzy objects." *ACM Trans. Graphics*, 2(2), 91–108.
- Schutte, J. F., and Groenword, A. A. (2005). "A study of global optimization using particle swarms." *J. Global Optim.*, 31, 93–108.
- Shi, Y., and Eberhart, R. C. (1998). "Parameter selection in particle swarm optimization." *Proc., 7th Int. Conf. on Evolutionary Programming*, Vol. 7, 591–600.
- Sivanesarwan, N., Kramer, S. L., and Mahoney, J. P. (1991). "Advanced back-calculation using a non-linear least squares optimization technique." *Transp. Res. Rec.*, 1293, 93–102.
- Terzi, S., Saltan, M., and Yildirim, T. (2003). "Optimization of deflection basin by genetic algorithm and neural network approach." *Proc., Int. Conf. on Neural Networks (ICANN/ICONIP)*, 662–669.
- Thompson, M. R. (1987). "ILLI-PAVE based full-depth asphalt concrete pavement design procedure." *Proc., 6th Int. Conf. on Structural Design of Asphalt Pavements*, Ann Arbor, Mich.
- Thompson, M. R., and Elliot, R. P. (1985). "ILLI-PAVE based response algorithms for design of conventional flexible pavements." *Transp. Res. Rec.*, 1043, 50–57.
- Thompson, M. R., and Robnett, Q. L. (1979). "Resilient properties of subgrade soils." *Transp. Engrg. J.*, 105(TE1), 71–89.
- Tsoukalas, L. H., and Uhrig, R. E. (1997). *Fuzzy and neural approaches in engineering*, Wiley, New York.
- Uzan, J. (1994). "Advanced backcalculation techniques." *NDT of pavements and back-calculation of moduli*, H. L. Von Quintus, A. J. Bush, and G. Y. Baladi, eds., Vol. 2, ASTM, West Conshohocken, Pa., 3–37.
- Van Cauwelaert, F. J., Alexander, D. R., White, T. D., and Barker, W. R. (1989). "Multilayer elastic program for elastic program for backcalculating layer moduli in pavement evaluation." *Nondestructive testing of pavements and backcalculation of moduli*, ASTM, West Conshohocken, Pa.